

CS 4727/5727 – Adversarial Machine Learning

Semester: Spring 2026 (January 14 – May 15, 2026)

Credit Hours: 3

Instructor: Alex Vakanski, vakanski@uidaho.edu

Office Location: TAB 311, Idaho Falls Center

Office Hours: Friday 12 p.m. – 1 p.m. PT (Zoom link available on Canvas)

Course Delivery Methods

- Virtual meetings (live meetings, students participate through Zoom)
 - Classroom (live meetings, video link from Idaho Falls)
 - Online (recorded Zoom videos of lectures available to students to watch after the classes)
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Course Description

The course introduces students to adversarial attacks on machine learning models and defenses against the attacks. The particular focus is on adversarial attacks and adversarial examples in deep learning models, due to their prevalence in modern machine learning applications. Covered topics include evasion attacks targeting white-box and black-box machine learning models, data poisoning attacks, privacy attacks, jailbreak attacks on large language models, defense strategies against adversarial attacks, and robust machine learning models. The course also provides an overview of adversarial attacks on machine learning models used in cybersecurity applications, including malware detection and classification, network intrusion detection, spam filtering, URL detection, cyber-physical systems, and biometric systems. Additional work is required for graduate credits.

Learning Outcomes

The objective is that upon the completion of the course the students should demonstrate the ability to:

1. Identify the vulnerabilities of machine learning models to various types of adversarial attacks.
2. Differentiate between adversarial evasion attacks in white-box and black-box settings and understand the principles of data poisoning attacks.
3. Explain the fundamentals of adversarial privacy attacks and outline privacy-preserving defense methods.
4. Describe jailbreak and poisoning attacks on large language models and propose corresponding mitigation methods.
5. List common defense strategies against adversarial attacks and discuss approaches for improved robustness of machine learning models.
6. Identify the unique characteristics of adversarial attacks on machine learning models in the cybersecurity domain.
7. Implement adversarial attacks and defenses against conventional machine learning models and deep learning models.
8. Evaluate the effectiveness of adversarial attacks on anomaly detection systems for network intrusion detection, machine learning malware classifiers, and anti-spam filtering models.
9. Analyze the ethical and societal implications of adversarial attacks and defenses.

Prerequisites

CS 2120 Practical Python, or CS 4771 Python for Machine Learning, or Instructor's Permission.

Students are expected to have a basic understanding of linear algebra, probability and statistics, and machine learning concepts. Familiarity with neural networks and deep learning is recommended, but not required. A working knowledge of Python programming is required to complete the course assignments and the course project. Additionally, it is preferred that the students have a basic level of familiarity with at least one of the following machine learning libraries: TensorFlow, Keras, or PyTorch.

Grading

Five homework assignments (worth together 30 marks), four quizzes (40 marks), course project (10 marks), presentation (10 marks), and class attendance and participation (10 marks). The assignments and the project will involve implementation of the studied adversarial attacks and defense methods on medium-size datasets (e.g., a few thousand images or similar). To complete the assignments and the project, the students can either use the GPUs provided by Google Colab, or they can use other programming environments that provide access to GPUs.

<i>Assessment Component</i>	<i>Marks</i>
Assignments (x5)	30
Quizzes (x4)	40
Course Project	10
Presentation	10
Attendance and Participation	10
<i>Total</i>	<i>100</i>

Textbooks

There are no required textbooks for this course.

Course Outline (Tentative)

Date	Topics, Readings, Assignments
Week 1	
Thursday, January 15	Lecture 1: Introduction to Adversarial Machine Learning
Week 2	
Tuesday, January 20	Lecture 1 (cont'd): Introduction to Adversarial Machine Learning
Thursday, January 22	Lecture 2: Deep Learning Overview
Week 3	
Tuesday, January 27	Lecture 2 (cont'd): Deep Learning Overview
Thursday, January 29	Lecture 3: Mathematics for Machine Learning <i>Reading:</i> Goodfellow (2014) Explaining and Harnessing Adversarial Examples (pdf) Due: Assignment 1
Week 4	

Tuesday, February 3	Lecture 3 (cont'd): Mathematics for Machine Learning
Thursday, February 5	Lecture 4: White-box Evasion Attacks 1. Carlini (2017) Towards Evaluating the Robustness of Neural Networks (pdf) <i>Optional readings:</i> 1. Papernot (2016) The Limitations of Deep Learning in Adversarial Settings (pdf) 2. Xiao (2018) Spatially Transformed Adversarial Examples (pdf) 3. Su (2019) One Pixel Attack for Fooling Deep Neural Networks (pdf) Quiz 1
Week 5	
Tuesday, February 10	Lecture 4 (cont'd): White-box Evasion Attacks
Thursday, February 12	Lecture 5: Black-box Evasion Attacks 1. Brendel (2017) Decision-Based Adversarial Attacks: Reliable Attacks Against Black-Box Machine Learning Models (pdf) <i>Optional readings:</i> 1. Papernot (2016) Transferability in Machine Learning: From Phenomena to Black-Box Attacks using Adversarial Samples (pdf) 2. Chen (2019) HopSkipJumpAttack: A Query-efficient Decision-based Adversarial Attack (pdf) 3. Guo (2019) Simple Black-box Adversarial Attacks (pdf)
Week 6	
Tuesday, February 17	Lecture 5 (cont'd): Black-box Evasion Attacks
Thursday, February 19	Lecture 6: Introduction to Large Language Models
Week 7	
Tuesday, February 24	Lecture 7: Jailbreak Attacks on Large Language Models 1. Zou (2023) Universal and Transferable Adversarial Attacks on Aligned Language Models (pdf) <i>Optional readings:</i> 1. Paulas (2024) AdvPrompter: Fast Adaptive Adversarial Prompting for LLMs (pdf) 2. Wei (2023) Jailbroken: How Does LLM Safety Training Fail? (pdf) 3. Qi (2024) Visual Adversarial Examples Jailbreak Aligned Large Language Models (pdf) 4. Ding (2023) A Wolf in Sheep's Clothing: Generalized Nested Jailbreak Prompts Can Fool Large Language Models Easily (pdf) Due: Assignment 2
Thursday, February 26	Lecture 7 (cont'd): Prompt Injection Attacks on Large Language Models
Week 8	
Tuesday, March 3	Lecture 8: Defenses Against Evasion Attacks 1. Cohen (2019) Certified Adversarial Robustness via Randomized Smoothing (pdf)

	<i>Optional readings:</i> 1. Madry (2017) Towards Deep Learning Models Resistant to Adversarial Attacks (pdf) 2. Zhang (2019) Theoretically Principled Trade-off between Robustness and Accuracy (pdf) 3. Carlini (2022) (Certified!!) Adversarial Robustness for Free! (pdf) 4. Zou (2024) Improving Alignment and Robustness with Circuit Breakers (pdf)
Thursday, March 5	Lecture 8 (cont'd): Defenses Against Evasion Attacks Due: Assignment 3
Week 9	
Tuesday, March 10	<u>Quiz 2</u>
Week 10	
Tuesday, March 24	Lecture 9: Poisoning Attacks 1. Liu (2018) Trojaning Attack on Neural Networks (pdf) <i>Optional readings:</i> 1. Souri (2022) Sleeper Agent: Scalable Hidden Trigger Backdoors for Neural Networks Trained from Scratch (pdf) 2. Souly (2025) Poisoning Attacks on LLMs Require a Near-Constant Number of Poison Samples (pdf) 3. Wan (2023) Poisoning Language Models During Instruction Tuning (pdf)
Thursday, March 26	Lecture 10: Defenses Against Poisoning Attacks 1. Wang (2019) Neural Cleanse: Identifying and Mitigating Backdoor Attacks in Neural Networks (pdf) <i>Optional readings:</i> 1. Huang (2019) NeuronInspect: Detecting Backdoors in Neural Networks via Output Explanations (pdf) 2. Steihhardt (2017) Certified Defenses for Data Poisoning Attacks (pdf)
Week 11	
Tuesday, March 31	Lecture 11: Adversarial Machine Learning in Cybersecurity – Part I, Network Intrusion Detection 1. Severi (2021) Explanation-Guided Backdoor Poisoning Attacks Against Malware Classifiers (pdf) <i>Optional readings:</i> 1. Rosenberg (2021) Adversarial Machine Learning Attacks and Defense Methods in the Cyber Security Domain (pdf) 2. Kuppa (2019) Black Box Attacks on Deep Anomaly Detectors (pdf) 3. Demetrio (2019) Explaining Vulnerabilities of Deep Learning to Adversarial Malware Binaries (pdf)
Thursday, April 2	Lecture 11 (cont'd): Adversarial Machine Learning in Cybersecurity – Part II, Malware Detection Due: Assignment 4
Week 12	
Tuesday, April 7	Lecture 11 (cont'd): Adversarial Machine Learning in Cybersecurity – Part III, Spam Filtering and URL Detection

	1. Erba (2019) Constrained Concealment Attacks against Reconstruction-based Anomaly Detectors in Industrial Control Systems (pdf) <i>Optional readings:</i> 1. Shirazi (2019) Adversarial Sampling Attacks Against Phishing Detection (pdf) 2. Anderson (2016) DeepDGA: Adversarially-Tuned Domain Generation and Detection (pdf) 3. Wang (2018) With Great Training Comes Great Vulnerability: Practical Attacks against Transfer Learning (pdf)
Thursday, April 9	Lecture 11 (cont'd): Adversarial Machine Learning in Cybersecurity – Part IV, Cyber-physical and Biometric Systems
Week 13	
Tuesday, April 14	Lecture 12: Privacy Attacks in Machine Learning 1. Shokri (2018) Membership Inference Attacks Against Machine Learning Models (pdf) <i>Optional readings:</i> 1. Rigaki (2021) A Survey of Privacy Attacks in Machine Learning (pdf) 2. Fredrikson (2015) Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures (pdf)
Thursday, April 16	Lecture 12 (cont'd): Privacy Attacks in Machine Learning <u>Quiz 3</u>
Week 14	
Tuesday, April 21	Lecture 13: Defenses Against Privacy Attacks 1. Abadi (2016) Deep Learning with Differential Privacy (pdf) <i>Optional readings:</i> 1. Nasr (2018) Machine Learning with Membership Privacy using Adversarial Regularization (pdf) 2. Papernot (2018) Scalable Private Learning with PATE (pdf) 3. McKenna (2025) Scaling Laws for Differentially Private Language Models (pdf) Due: Assignment 5
Thursday, April 23	Lecture 13 (cont'd): Defenses Against Privacy Attacks Due: <u>Project proposal</u>
Week 15	
Tuesday, April 28	Lecture 14: Explainability in Machine Learning 1. Templeton (2024) Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet (pdf) <i>Optional readings:</i> 1. Cunningham (2023) Sparse Autoencoders Find Highly Interpretable Features in Language Models (pdf) 2. Sundararajan (2017) Axiomatic Attribution for Deep Networks (pdf) 3. Montavon (2017) Explaining Nonlinear Classification Decisions with Deep Taylor Decomposition (pdf)
Thursday, April 30	Lecture 14 (cont'd): Explainability in Machine Learning

Week 16	
Tuesday, May 5	Lecture 15: Bias and Fairness in Machine Learning 1. Navigli (2023) Biases in Large Language Models: Origins, Inventory, and Discussion (pdf) <i>Optional readings:</i> 1. Gallegos (2023) Bias and Fairness in Large Language Models: A Survey (pdf) 2. Dai (2024) Bias and Unfairness in Information Retrieval Systems: New Challenges in the LLM Era (pdf) 3. Anthiis (2024) The Impossibility of Fair LLMs (pdf) 4. Ye (2024) Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge (pdf)
Thursday, May 7	Lecture 15 (cont'd): Bias and Fairness in Machine Learning Due: Project report
Week 17	
Tuesday, May 12	<u>Quiz 4</u>

Academic Integrity

Students are expected to adhere to the highest academic standards of honesty and integrity. At UI, we assume students will do their own work. Plagiarism—passing off someone else's work as your own, without citing the source—would not be tolerated. This includes direct copying, rephrasing, and summarizing, as well as taking someone else's idea and putting it in different words. The best avenue for avoiding plagiarism issues is to fully cite all sources used for preparing assignments, texts, and exams.

Learning Civility

In any environment in which people gather to learn, it is essential that all members feel as free and safe as possible in their participation. To this end, it is expected that everyone in this course will be treated with mutual respect and civility, with an understanding that all of us (students, instructors, professors, guests, and teaching assistants) will be respectful and civil to one another in discussion, in action, in teaching, and in learning.

Should you feel our classroom interactions do not reflect an environment of civility and respect, you are encouraged to meet with your instructor during office hours to discuss your concern. Additional resources for expression of concern or requesting support include the Dean of Students office and staff (208-885-6757), the UofI Counseling & Testing Center's confidential services (208-885-6716), or the Office of Civil Rights and Investigations (208-885-4285).

Center for Disability Access & Resources (CDAR)

University of Idaho is committed to ensuring an accessible learning environment where course or instructional content are usable by all students and faculty. If you believe that you require disability-related academic adjustments for this class (including pregnancy-related disabilities), please contact Center for Disability Access and Resources (CDAR) to discuss eligibility. A current accommodation letter from CDAR is required before any modifications, above and beyond what is otherwise available for all other students in this class will be provided. Please be advised that disability-related academic

adjustments are not retroactive. CDAR is located at the Bruce Pitman Building, Suite 127. Phone is 208-885-6307 and e-mail is cdar@uidaho.edu. For a complete listing of services and current business hours visit <https://www.uidaho.edu/current-students/cdar>.

Healthy Vandals Policies

Please visit the [University of Idaho COVID-19 webpage](#) often for the most up-to-date information about the UofI's response to Covid-19.

Vandal Food Pantry

The [Vandal Food Pantry](#) is a free resource stocked weekly with food, grocery bags, and various hygiene items. Its eight locations across campus are accessible during building hours and open to all. Please take what you need.

Green Dot Safety Program

What's Your Green Dot? It's up to all of us to make a safer campus. Vandal Green Dot is a program that helps students learn about the power of the bystander, how to recognize potentially risky situations, and realistic ways to intervene. Together we can bring down the number of people being hurt by interpersonal violence on our campus. No one has to do everything, but everyone has to do something! Learn more and get involved by visiting [UI's Green Dot Safety Program](#) or emailing greendot@uidaho.edu.

Help and Resources

Student Resources

The University of Idaho provides student support to ensure a successful learning experience.

- [Student Resources Webpage](#)
- [SI-PASS \(Peer Assisted Study Sessions\)](#) SI-PASS provides regularly scheduled, peer-led study sessions for difficult courses.

Library Help

The UofI Library website has many databases that will help you find relevant and reliable books, articles, images, and more. Don't hesitate to contact a librarian for research assistance.

- [UofI Library Website](#)
- [Help - Reference Services](#)
- [Help for Distance Ed Students](#)

Technology Help

The UofI Student Technology Center provides many technology-related services to students.

- PHONE: 208-885-HELP (208-885-4357)
- Technology Help Email: support@uidaho.edu
- [Technology Help Website](#)

Writing Support

The UofI Writing Center provides one-on-one assistance to student writers and other members of the campus community.

- PHONE: 208-885-6644
- Writing Center Email: writingcenter@uidaho.edu
- [Writing Center Website](#)

UofI Moscow Land Acknowledgement

UofI Moscow is located on the homelands of the Nimiipuu (Nez Perce), Palus (Palouse) and Schitsu'umsh (Coeur d'Alene) tribes. We extend gratitude to the indigenous people that call this place home, since time immemorial. UofI recognizes that it is our academic responsibility to build relationships with the indigenous people to ensure integrity of tribal voices.